

Session 10 Quiz - Strengths & Limits of Generative AI

Q1 [Type: MCQ | Difficulty: Easy | QID : 52977]

One common strength of Generative AI is _____, which is why a one-page club flyer that might take you an hour can be drafted in just seconds.

- A. judgment
- B. verification
- C. speed
- D. common sense

Correct Answer: C

Explanation: Speed is the first strength named in this session - AI can generate drafts, flyers, and outlines in seconds. Note that speed is not the same as accuracy; fast output still needs human review.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52979]

The _____ approach is a framework where AI makes suggestions, but humans make the final decisions and remain involved, checking, and responsible throughout the process.

- A. Autopilot Override
- B. Human-in-the-Loop
- C. AI-First Workflow
- D. Confidence Calibration

Correct Answer: B

Explanation: Human-in-the-Loop is the framework introduced this session: the human prompts, AI generates, the human reviews, and the human uses or refines the output. The human is always in the loop and remains responsible.

Q3 [Type: MCQ | Difficulty: Medium | QID : 53007]

When AI generates a list of "famous athletes" that is mostly men because its training data over-represented male athletes in media coverage, this is an example of _____.

- A. hallucination
- B. translation drift

C. bias

D. an outdated training cutoff

Correct Answer: C

Explanation: Bias means the AI reflects patterns and over-representations in its training data. It is not intentional, but it shapes the outputs you get unless you prompt specifically (for example, "famous female athletes" or "athletes from different countries").

Q4 [Type: MCQ | Difficulty: Easy | QID : 53009]

A chatbot is a reliable source for medical advice, such as recommending which painkiller a friend should take for a headache.

A. True

B. False

Correct Answer: B (False)

Explanation: AI does not know a person's age, allergies, current medications, or medical history. Health and safety are explicitly listed in the session as situations to avoid AI; ask a doctor, pharmacist, or trusted adult instead.

Q5 [Type: MCQ | Difficulty: Medium | QID : 53011]

When a chatbot correctly answers "a bowling ball hits the ground first" to a physics question, it shows that the AI truly understands gravity and air resistance.

A. True

B. False

Correct Answer: B (False)

Explanation: Generative AI mimics patterns from training data rather than comprehending meaning. A correct answer does not prove understanding - the same AI often cannot explain why the result holds or how it would change in different conditions (for example, on the moon).

Q6 [Type: MCQ | Difficulty: Medium | QID : 53012]

Different Generative AI tools (such as ChatGPT, Gemini, and NotebookLM) can have meaningfully different strengths and limits depending on how they are designed.

A. True

B. False

Correct Answer: A (True)

Explanation: NotebookLM, for example, refuses to answer without sources, while ChatGPT will almost always attempt an answer. Part of being a smart user is learning the specific strengths and limits of the tool you are actually using - not assuming every chatbot behaves the same.

Q7 [Type: MCQ | Difficulty: Easy | QID : 53013]

Which of the following is a common STRENGTH of Generative AI?

- A. Knowing your school's latest schedule updates from this week
- B. Generating multiple draft ideas quickly when you are stuck
- C. Making medical decisions for a sick friend
- D. Reliably reporting today's breaking news

Correct Answer: B

Explanation: Brainstorming and drafting are core strengths - AI is good at producing many options quickly. A, C, and D each describe situations the session explicitly warns about (outdated info, health decisions, time-sensitive facts).

Q8 [Type: MCQ | Difficulty: Easy | QID : 53015]

Which of the following is a common LIMIT of Generative AI?

- A. Producing creative variations of a draft
- B. Helping translate a short message into another language
- C. Confidently making up information
- D. Brainstorming names or titles

Correct Answer: C

Explanation: Hallucination - confidently generating false information - is a defining limit of Generative AI. The other options are all listed strengths in this session.

Q9 [Type: MCQ | Difficulty: Easy | QID : 53017]

Why is "Explaining Complex Ideas" listed as a strength of GenAI?

- A. AI only uses the most advanced vocabulary possible
- B. AI can rephrase information using helpful analogies
- C. AI has a PhD in most scientific subjects
- D. AI always provides accurate scientific proof

Correct Answer: B

Explanation: Its ability to relate difficult concepts (like electricity) to simple ones (like water flow) makes it a great tutor - even though the AI does not truly understand the underlying concept itself.

Q10 [Type: MCQ | Difficulty: Easy | QID : 53018]

How does NotebookLM differ from ChatGPT in its approach to providing information?

- A. NotebookLM refuses to answer if it cannot find the source
- B. There is no functional difference in how they handle sources
- C. NotebookLM always tries to answer even without data
- D. ChatGPT only uses information from the last 24 hours

Correct Answer: A

Explanation: NotebookLM is designed to be grounded in specific documents, reducing the likelihood of making things up. Saying "I don't know" is a feature, not a bug.

Q11 [Type: MCQ | Difficulty: Easy | QID : 53019]

In the Human-in-the-Loop approach, what is the primary role of the human after the AI generates an initial response?

- A. To blindly accept the result to save time
- B. To increase the AI's confidence score
- C. To delete the output and start over manually
- D. To review, refine, and take responsibility

Correct Answer: D

Explanation: The human must act as the supervisor, checking for accuracy and making the final decisions on how the information is used.

Q12 [Type: MCQ | Difficulty: Medium | QID : 53021]

Why does Generative AI sometimes produce biased outputs, such as listing only male athletes when asked for "famous athletes"?

- A. Because it is intentionally programmed to favor certain groups
- B. Because it reflects patterns from its training data
- C. Because it has personal opinions it tries to express
- D. Because it does not understand what the word "famous" means

Correct Answer: B

Explanation: Bias in AI outputs typically comes from biases in training data. Historically, male athletes received more media coverage, so a "famous athletes" prompt without further context inherits that skew. Specifying the prompt (e.g., adding "from different countries") helps balance the result.

Q13 [Type: MCQ | Difficulty: Medium | QID : 53023]

Using the "Pilot and Autopilot" analogy, what is the pilot's responsibility regarding the flight route?

- A. Letting the autopilot handle the flight without supervision
- B. Allowing the autopilot to change the route based on its own logic
- C. Assuming the autopilot will correct its own errors
- D. Programming the correct route and monitoring decisions

Correct Answer: D

Explanation: The human (pilot) sets the goals and must be ready to take over immediately if the AI (autopilot) makes an error.

Q14 [Type: MCQ | Difficulty: Medium | QID : 53028]

You ask AI for the current score of a basketball game happening right now. Why is this a "NOT to use" situation?

- A. AI has a training data cutoff and lacks real-time awareness
- B. AI struggles with numbers and sports statistics
- C. Sports scores are too complex for AI to process
- D. AI is biased against certain sports teams

Correct Answer: A

Explanation: Most AI tools do not have access to live, up-to-the-minute information unless specifically connected to a search engine.

Q15 [Type: MCQ | Difficulty: Hard | QID : 53029]

A classmate argues, "If the AI sounds confident, the answer is probably right." Based on this session, what is the best response?

- A. Agree - confidence almost always tracks with accuracy in modern AI
- B. Disagree - AI can be confidently wrong
- C. Agree, but only when the AI uses formal language
- D. Disagree only for math; otherwise confidence is reliable

Correct Answer: B

Explanation: Confidence $\neq$ accuracy. Hallucinations like the Riverside Park / Frederick Olmsted example are dangerous precisely because they sound authoritative. Verification is what separates an answer from a guess.

Q16 [Type: MCQ | Difficulty: Hard | QID : 53031]

Which of these is the most responsible "Personal Rule" for using AI based on the session's teachings?

- A. If it sounds confident, it is probably correct
- B. Only use AI for tasks that take more than three hours
- C. Trust the AI to make personal choices about friendships
- D. Always be the "pilot" and verify important information

Correct Answer: D

Explanation: Maintaining responsibility and checking the output ensures the AI supports rather than replaces your thinking.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 53034]

Scenario: You ask AI about the history of your town's oldest park. It tells you, "Riverside Park was created in 1895 by famous landscape architect Frederick Olmsted, and the iron bridge was added in 1908." You then check with the local historical society and learn the park was actually created in 1910, has no connection to Olmsted, and the bridge was built in 1925.

Question: Which limit of Generative AI does this best illustrate?

- A. Bias
- B. Translation error
- C. Hallucination
- D. Outdated training data

Correct Answer: C

Explanation: This is a textbook hallucination: highly specific, authoritative-sounding details (a date, a famous name, an iron bridge year) that are simply not true. The right move after spotting it is to verify with a trustworthy source - exactly what you did with the historical society.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 53037]

Scenario: Your class is planning a "Kindness Assembly." One student does the following:

- (1) Prompts AI for 10 assembly ideas.

- (2) Reviews the list and removes ideas that exceed the school's budget.
- (3) Prompts AI again, narrowing the request to student performance ideas.
- (4) Forms committees with classmates to actually carry out the chosen ideas.

Question: Which choice best describes what this student is doing?

- A.** Replacing human decision-making with AI to save time
- B.** Following the Human-in-the-Loop approach
- C.** Outsourcing the assembly to AI, which is responsible for the outcome
- D.** Using AI only as a translator between languages

Correct Answer: B

Explanation: Every step the student takes - prompting, reviewing, refining the prompt, and finally deciding with classmates - keeps the human in the loop. The AI assists; the student directs and remains responsible for the final plan.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 53039]

Scenario: A student in your class wants to write a speech for an end-of-year ceremony. She tells you she plans to type the topic into a chatbot, copy whatever speech it produces, and read it aloud the next day without changes.

Question: Using the Human-in-the-Loop approach, explain what is missing from her plan and describe what a better workflow would look like.

(Respond in 3–4 sentences.)

Model Answer: Her plan removes the human from the loop because she is treating the AI's draft as the final product instead of as a starting point. A better workflow would prompt the AI for ideas or a rough draft, then review it carefully for accuracy and tone, edit it to reflect her own experiences and voice, and verify any specific facts, names, or quotes it includes. Under Human-in-the-Loop, the AI assists her thinking, but she directs, refines, and remains responsible for the final speech. That way the audience hears something authentic to her rather than a polished but generic - and possibly inaccurate - chatbot output.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53040]

Scenario: Your friend is upset after an argument with a classmate and is unsure whether to apologize. He suggests asking a chatbot, "Should I apologize to my friend?" so the AI can make the decision for him.

Question: Should he use AI to make this decision? Justify your answer using at least two concepts from this session (for example: common limits, when to avoid AI, or Human-in-the-Loop).

(Respond in 3–4 sentences.)

Model Answer: He should not let AI make the decision for him. Personal and emotional decisions are listed as "avoid AI" situations because the chatbot does not know his friendship, the tone of the argument, or the people involved - and AI in general has no real understanding of human relationships, only patterns from text. He could still use AI in a Human-in-the-Loop way, for example to brainstorm how to phrase an apology if he chooses to make one, but the decision of whether to apologize must stay with him because he is the one responsible for the friendship. Trusting his own judgment, or talking with a trusted friend or adult, is the right move here.